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Salivary microbial diversity at different stages of human immunodeficiency virus infection

Ying Guo [a,b], Wei Xia [c], Feili Wei [d], Wei Feng [c], Junyi Duan [c], Xintong Sun [b], Yixi Yu [b], Yan Wang [b], Baojin Zhang [b], Jie Cao [b], Weidi Li [b], Haozhi Dong [e], Shan Guo [d], Fengqiu Zhang [a,***], Xiaojie Huang [c,**], Zheng Sun [a,*]

[a] Capital Medical University School of Stomatology, Beijing, 100050, China
[b] Department of Stomatology, Beijing Youan Hospital, Capital Medical University, Beijing, 100069, China
[c] Department of Infectious Disease, Beijing Youan Hospital, Capital Medical University, Beijing, 100069, China
[d] Beijing Institute of Hepatology, Beijing Youan Hospital, Capital Medical University, Beijing, 100069, China
[e] Department of Stomatology, Beijing Daxing District Hospital of Integrated Chinese and Western Medicine, Beijing, 100076, China

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ABSTRACT

Human immunodeficiency virus (HIV)/acquired immunodeficiency syndrome (AIDS) disrupts the host microbial balance. During disease progression, the oral microbial environment is altered in untreated people living with HIV/AIDS (PLWHA); however, no studies have reported changes in salivary microbial diversity during different stages of HIV infection. Therefore, in this study, we aimed to assess the relationships between immune dysfunction and changes in saliva microbiota. To this end, we collected saliva samples from 11 HIV-negative individuals and 44 PLWHA during different stages based on the Centers for Disease Control and Prevention criteria (stage 0, early stage during the first 6 months after infection; stages 1, 2, and 3 associated with $CD4^+$ T-lymphocyte counts of $\geq$500, 200–499, and $\leq$200 or opportunistic infection, respectively). We analyzed salivary microbial community diversity using polymerase chain reaction amplification and Illumina MiSeq sequencing. We found that HIV-positive individuals had significantly greater alpha-diversity in the microbial community composition compared with HIV-negative controls ($P < 0.05$) except for AIDS (stage 3); however, the predominant salivary microbiota in the five groups remained similar. *Porphyromonas* in the four positive groups was the only genus that was significantly less abundant in the HIV-positive groups than in the control group ($P < 0.05$). There were some consistencies between the general abundance of salivary microbiota and AIDS disease progression. Lots of bacterial abundances in the saliva increased dramatically during the acute HIV infection (stage 0), and some of the negligible and abnormally proliferating bacteria in the asymptomatic stage showed a downward trend. Additionally, in the AIDS stage, partial inhibition was observed. Notably, *Porphyromonas* was closely related to the immune activation of HIV, showing a decline in abundance once infected with HIV. *Solobacterium*, which induces inflammation, was negatively correlated with CD4 counts. Overall, our findings provided important insights into changes in salivary microbial diversity in PLWHA.

1. Introduction

The oral microbiota is a dynamic community that interacts with the host to drive health or disease through homeostasis or dysbiosis [1]. When appropriately coordinated, the balance of the host's immune system and oral microbiota constitutes a healthy oral environment. At the end of 2019, there were 38 million people living with human immunodeficiency virus (HIV)/acquired immunodeficiency syndrome (AIDS; PLWHA) globally, 18.7% of whom are not aware of their status [2]. Oral lesions are often the first symptom for identifying PLWHA [3]. When HIV invades the body and spreads from the original site of infection, the disease enters the period of acute HIV infection (AHI),

* Corresponding author. Capital Medical University School of Stomatology, Beijing, China.
** Corresponding author. Department of Infectious Disease, Beijing Youan Hospital, Capital Medical University, Beijing, China.
*** Corresponding author. Capital Medical University School of Stomatology, Beijing, China.
E-mail addresses: zhfengqiu@126.com (F. Zhang), huangxiaojie78@ccmu.edu.cn (X. Huang), sunzheng12@vip126.com (Z. Sun).

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characterized by a high viral load and infectiousness [4]. The balance of the host microbiota begins to be disrupted, and the host displays inflammation associated with microbial dysbiosis. Pharyngitis is usually the first manifestation of oral inflammation [5–7]. After AHI, patients enter an asymptomatic stage. Although patients may have no oral lesions, oral mucosal tissues can exhibit a similar pattern of gastrointestinal disruption owing to the chronic status of increased immune activation and inflammation, inducing microbial translocations and changes [8–10]. Once the disease progresses to have a CD4 count of less than 200 positive cells/μL, called the AIDS stage, the host is highly susceptible to oral opportunistic infection and malignant tumors, including oral candidiasis, linear erythema, Kaposi's sarcoma, or non-Hodgkin's lymphoma [11]. These oral manifestations may be closely related to oral microbiome imbalance [9,12,13]. In AIDS patients with severe immunosuppression, these oral infections highlight the importance of host/microbe interactions [14]. Therefore, it is very necessary to understand the oral microbiota under different HIV disease courses.

Saliva is a biological fluid found in the oral cavity that contains bacteria emitted from the surface of oral tissue. The oral microbiota in a planktonic state can be transmitted outside the body through saliva as a fluid medium [15,16]. Therefore, saliva is an important research vehicle for evaluation of the oral microbiota of the host. In most of recent studies on saliva, patients with HIV have not been clearly distinguished according to whether they had received antiretroviral therapy (ART) [17–19]. Alternatively, some studies only chose patients after receiving ART [20–23], and only two studies had focused on saliva microbes before receiving ART, one found that the microbial diversity of saliva was higher in the HIV-positive group than in the HIV-negative group, whereas another study showed the opposite results [24,25]. Thus, further studies, including research on the oral microbiota of HIV-positive patients with different disease stages, are needed. Understanding changes to the oral microbiota during different stages of the disease could guide the prevention of HIV in patients who have not developed oral disease.

Therefore, in this study, we focused on assessing the oral environment among PLWHA who had not received ART. We aimed to explore changes in the oral microbiota during HIV infection and to assess relationships between immune dysfunction and the HIV-associated microbiota by comparing the salivary microbiota among patients with different immunosuppressive status.

2. Materials and methods

2.1. Study participants

We conducted a cross-sectional study of HIV-negative controls and untreated PLWHA. According to the staging criteria of the Centers for Disease Control and Prevention for monitoring cases [26], stage 0 is the early stage of HIV infection, i.e., seroconversion within 6 months or indeterminate results by western blotting; and stages 1, 2, and 3 are based on $CD4^+$ T-lymphocyte counts of greater than or equal to 500, 200–499, and less than 200 cells/μL or opportunistic infections, respectively. The study protocol was registered at Chinese Clinical Trial Registry (ChiCTR2000030301).

In total, 106 participants were randomly selected from the Infectious Diseases Center of Beijing Youan Hospital, Capital Medical University. All participants were Chinese men who have sex with men (MSM), lived in Beijing (China), were greater than 18 years old, and had received a full oral clinical examination. Those who received antibiotics, ART, and immunomodulatory drugs within the last 3 months; had a history of any systemic disease; had oral problems (excluding non-cavitated caries, nonpurulent periodontal disease or oral candidiasis); or had fewer than 20 teeth were excluded from the study. Fifty-five subjects were screened strictly according to the inclusion and exclusion criteria and divided into five groups according to HIV stage/status (HIV-positive individuals: stage 0 group [n = 11], stage 1 group [n = 10], stage 2 group [n = 13], stage 3 group [n = 10]; and HIV-negative control group [n = 11]).

2.2. Sample collection

The participants were instructed to collect saliva from 10:00 to 13:00 and refrain from eating, drinking, or oral hygiene procedures for at least 2 h prior to collection. We asked the participants to spit in to a 20-mL sterile tube, and we then placed the tube on ice in a polystyrene plastic box after collecting approximately 5 mL saliva. Samples were delivered to the laboratory within 1 h and stored at −80 °C until DNA extraction [27].

2.3. DNA extraction, polymerase chain reaction (PCR) amplification, and sequencing

Microbial community genomic DNA was extracted from saliva samples using a FastDNA SPIN Kit for soil (MP Biomedicals, Santa Ana, CA, USA) according to the manufacturer's instructions. We amplified the hypervariable region V3–V4 of the bacterial 16S rRNA gene with primer pairs 338F (5′-ACTCCTACGGGAGGCAGCAG-3′) and 806R (5′-GGACTACHVGGGTWTCTAAT-3′) with an ABI GeneAmp 9700 PCR thermocycler (ABI, CA, USA). PCR products were identified by 2% agarose gel electrophoresis. PCR amplification of the 16S rRNA gene was performed as follows: initial denaturation at 95 °C for 3 min; followed by 27 cycles of denaturation at 95 °C for 30 s, annealing at 55 °C for 30 s, and extension at 72 °C for 45 s; and a single extension at 72 °C for 10 min. After purification and quantification, amplicons were pooled at equimolar amounts and paired-end sequenced (2 × 300) on an Illumina MiSeq platform (Illumina, San Diego, CA, USA), according to standard protocols, by Majorbio Bio-Pharm Technology Co. Ltd. (Shanghai, China). The raw reads were deposited into the NCBI Sequence Read Archive database (accession number: SRP251412). The raw reads were demultiplexed, quality filtered by Trimmomatic, and assembled by FLASH. Operational taxonomic units (OTUs) used UPARSE (version 7.1, http://drive5.com/uparse/) for clustering, and the similarity cutoff was 97%. The taxonomy of each OTU representative sequence was analyzed by RDP Classifier (http://rdp.cme.msu.edu/) against the 16S rRNA database (e.g., Silva SSU132) using a confidence threshold of 0.7.

2.4. Statistical analysis

Statistical analysis was performed using SAS version 9.4 (SAS Institute Inc., Cary, NC, USA). Participants' characteristics were presented as medians and interquartile ranges. Kruskal-Wallis tests were used for comparisons between groups. Classification data were described in the form of numbers (percentages), and chi-square tests were used for comparisons between groups. Results with P values of less than 0.05 were considered statistically significant. Alpha-diversity was evaluated by the Sobs and Shannon indexes at the OTU level, and comparisons of differences between groups were made using Wilcoxon rank sum tests. Beta-diversity analysis was performed using partial least squares discriminant analysis (PLS-DA), which ignored random differences within groups and highlighted systematic differences between groups to reveal similarities between groups. Species difference analysis based on community abundance data used one-way analysis of variance to analyze differences in the species distributions of the five sample groups, performed using Tukey-Kramer tests (significance level: 0.95) to compare pairs. Finally, species with significant differences between groups at the genus level were identified. Correlation heatmaps were calculated through Spearman rank correlation analysis to determine the relationships of the top 20 species with CD4 counts and blood viral load (BVL).

Table 1
Characteristics of participants in untreated PLWHA and control groups.

Group	HIV_0 (n = 11)	HIV_1 (n = 10)	HIV_2 (n = 13)	HIV_3 (n = 10)	HIV_neg (n = 11)	Chi-square value	*P* value
Age, years	31.00 (24.00,41.00)	26.50 (24.75,34.25)	30.00 (25.50,42.00)	27.50 (24.50,34.25)	29.00 (26.00,30.00)	1.3184	0.8583
Periodontal state,n (%)							
Periodontal health	1 (9.09)	3 (30.00)	3 (23.08)	3 (30.00)	2 (18.18)	2.093	0.7187
Periodontitis	10 (90.91)	7 (70.00)	10 (76.92)	7 (70.00)	9 (81.82)		
Mucosal state,n (%)							
Mucosal health	11 (100.00)	10 (100.00)	13 (100.00)	7 (70.00)	11 (100.00)	11.068	0.0258
oral candidiasis	0 (0.00)	0 (0.00)	0 (0.00)	3 (30.00)	0 (0.00)		
CD4 counts, cells/μL	373.49 (328.00,447.30)	637.86 (550.51,680.85)	334.00 (240.50,405.00)	73.53 (28.46,198.47)	ND	34.1131	<0.0001
BVL, $\log_{10}$ copies/mL	4.28 (3.92, 5.06)	3.80 (3.42, 4.67)	4.23 (3.61, 5.13)	4.66 (4.03, 5.11)	0 (0, 0)	4.4102	0.2204

Abbreviations: BVL = blood viral load; ND=Not detected; HIV_0 = HIV-positive stage 0 group, HIV_1 = HIV-positive stage 1 group, HIV_2 = HIV-positive stage 2 group, HIV_3 = HIV-positive stage 3 group, HIV_neg = HIV-negative control group.
Characteristics of participants are summarized using means with upper and lower quartiles or percentages and compared using Chi-square tests.

Table 2
Alpha diversity analysis at the OTU level of five groups in the saliva microbiota study.

	HIV_0	HIV_1	HIV_2	HIV_3	HIV_neg	*P* value				
						0-neg	1-neg	2-neg	3-neg	2–3
Sobs (mean ± SD)	456.27 ± 156.04	439.2 ± 108.78	438.08 ± 73.146	377.4 ± 120.18	344.82 ± 59.871	0.04173*	0.04094*	0.006434**	0.751	0.04714*
Shannon (mean ± SD)	3.7584 ± 0.37103	3.6686 ± 0.38304	3.7686 ± 0.37453	3.5548 ± 0.31012	3.3876 ± 0.35013	0.01808*	0.02654*	0.00377**	0.4597	0.1003

Statistical analysis was performed by Wilcoxon rank sum tests; *$0.01 < P \leq 0.05$, **$0.001 < P \leq 0.01$.

3. Results

3.1. Participants' characteristics

Demographic details of the MSM participating in this study are shown in Table 1. Saliva samples from 11 HIV-negative controls and 44 HIV-positive patients were obtained. There were no significant differences in ages and periodontal states between the five groups ($P > 0.05$). Table 1 shows CD4 counts and BVL in each group of HIV-positive individuals. Notably, from stage 1 to 2 and 3, CD4 counts continued to decrease, and BVL gradually increased in untreated HIV-positive patients.

3.2. Comparison of microbial diversity in HIV-positive and HIV-negative control groups

Based on our analysis of 1985 16S rRNA OTUs, microbial diversity was detected in salivary samples. HIV-positive and -negative patients had 305 different microbial genera, representing 22 different microbial phyla.

Alpha-diversity in saliva from HIV-positive patients was significantly different compared with that in HIV-negative controls (Table 2). At the OTU level, Sobs analysis indicated that, except for stage 3, saliva from HIV-positive patients was significantly more abundant than that from HIV-negative controls. The richness of saliva samples in stage 2 was significantly higher than in stage 3 and the negative control groups, whereas those in stage 3 and the negative controls were similar (Fig. 1A). Furthermore, Shannon alpha-diversity analysis showed that saliva from HIV-positive patients, except for those with stage 3 group, was significantly more diverse compared with saliva from negative controls.(Fig. 1B).

Beta-diversity analysis showed that the microbial communities in the HIV-negative control group were clearly separated from those in the HIV-positive groups. As the CD4 counts decreased, microbiota at the genus level gradually migrated from stage 0 to stage 3 and deviated from the HIV-negative control group (Fig. 1C).

3.3. Comparison of microbial distributions in HIV-positive and HIV-negative groups

Venn diagram analysis was performed to distinguish the differences among the five groups. At the genus level, the number of species in the HIV-positive group was higher than that in the HIV-negative control group. There were 129 genera shared between all five groups, and each group had its own specific genera. The 22 and 26 most unique genera were detected in the stage 0 and 1 groups, whereas 15 unique genera were detected in the HIV-negative group (Fig. 2A).

There were 305 genera detected among the five groups. The predominant salivary microbiota in the five groups were similar and included *Streptococcus*, *Prevotella_7*, *Porphyromonas*, *Veillonella*, *Actinomyces*, *Neisseria*, *Prevotella* and *Rothia*. We visualized the relative abundances of the dominant taxa at the genus level in the salivary microbiota. The abundance of *Streptococcus* was the highest among all groups, with no significant differences between groups (Fig. 2B).

As shown in Fig. 3A–D showed that the genera were significantly different between the four HIV-positive groups and the HIV-negative control group. Compared with the HIV-negative control group, the abundances of eight bacterial genera, including *Porphyromonas* and *Rothia*, were reduced in the stage 0 group, whereas the abundances of 34 other genera, such as *Veillonella*, *Fusobacterium*, and *Alloprevotella*, increased. In the stage 1 group, the abundances of seven genera, including *Porphyromonas* and *Lactobacillus*, decreased, whereas the abundances of 15 genera, including *Megasphaera* and *Corynebacterium*, increased. For the stage 2 group, the abundances of eight genera, including *Porphyromonas* and *Prevotella*, decreased, as opposed to the increasing abundance of the bacteria such as *Veillonella*, *Alloprevotella*, *Megasphaera*, *Aggregatibacters*, *Selenomonas_3*, and *Campylobacter*. In the stage 3 group, the differences between genera began to narrow; the abundances of *Porphyromonas*, *Rothia*, *Haemophilus*, *Prevotella_2*, and seven other genera decreased, whereas the abundances of *Prevotella_6*,

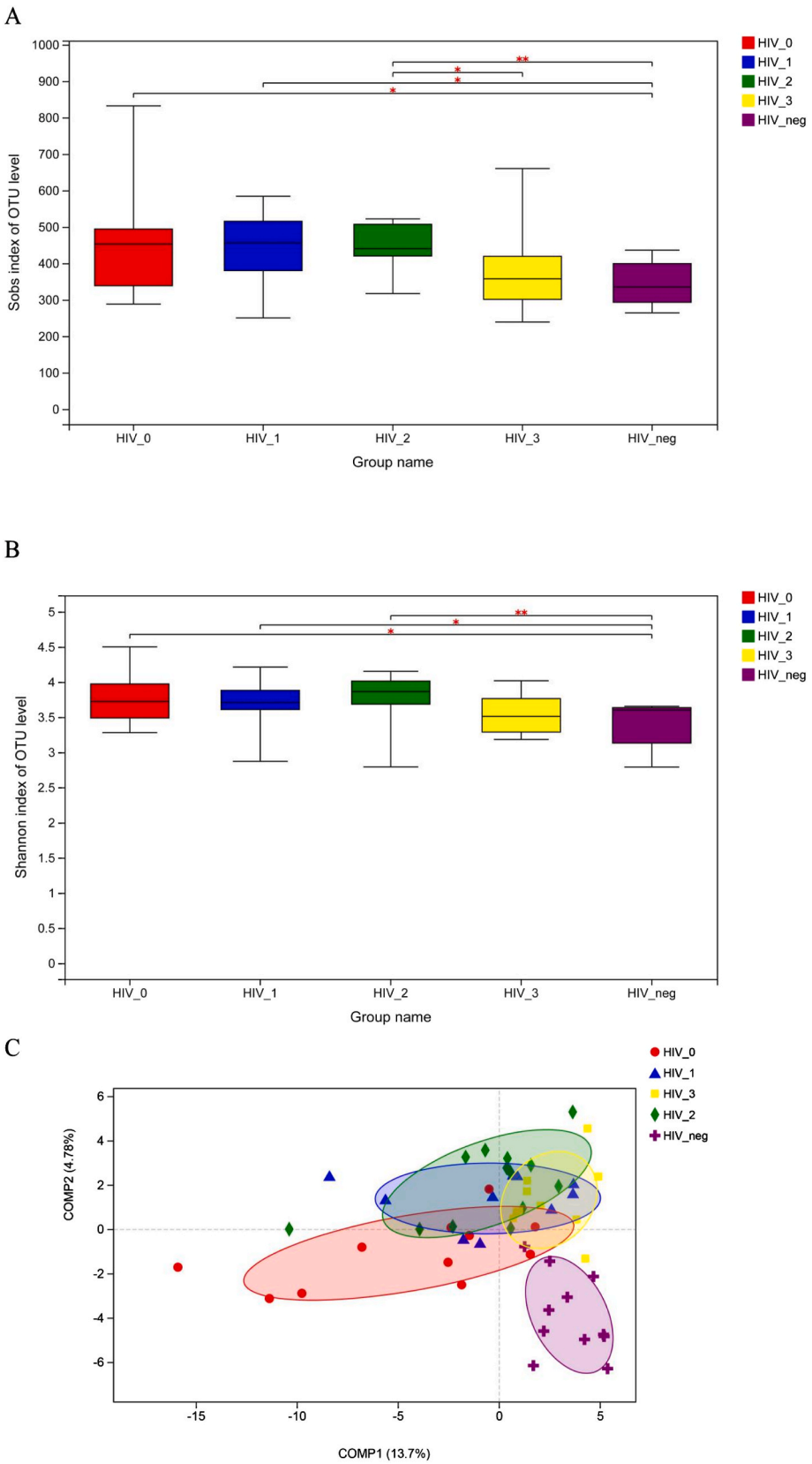


Fig. 1. Comparison of saliva microbial diversity in four HIV-positive groups and HIV-negative controls: stage 0 (red), stage 1 (blue), stage 2 (green), stage 3 (yellow), and controls (purple). *$0.01 < P \leq 0.05$, **$0.001 < P \leq 0.01$. A. Sobs index of the observed richness of alpha-diversity at the OTU level. B. Shannon index of the observed diversity of alpha-diversity at the OTU level. C. Beta-diversity comparisons by partial least squares discriminant analysis (PLS-DA).

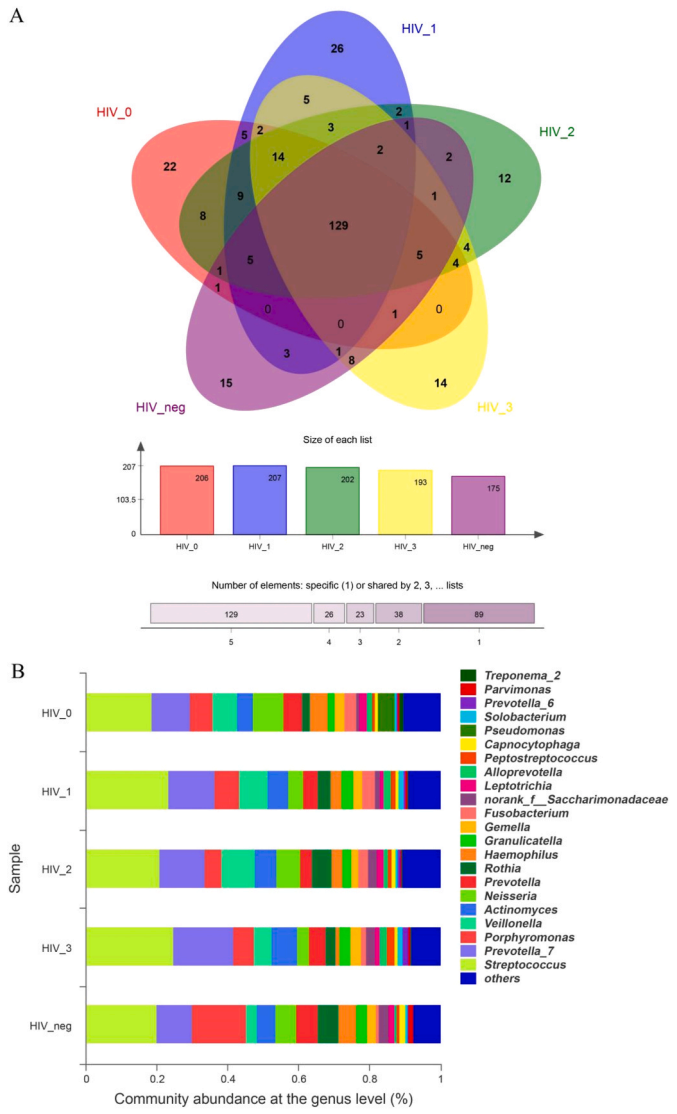


Fig. 2. Salivary community composition analysis at the genus level between HIV-positive groups and HIV-negative controls. A. The Venn diagram shows the number of species. The number in the overlapping part represents the number of species common between multiple groups, and the numbers in the nonoverlapping parts represent the number of species unique to the corresponding group. The bar chart shows the total number of species in each group. B. The community bar chart shows the species abundance in each group.

Campylobacter, *Shuttleworthia*, *Dialister*, and four other genera increased. A comparison of genus richness in all saliva samples demonstrated that changes in bacterial genera were the greatest in the stage 0 group and the smallest in the stage 3 group. Notably, *Porphyromonas* was significantly more abundant in controls than in the other groups and was negatively correlated with BVL (Coefficient: −0.024808, $P < 0.01$) (Fig. 4).

There were some consistencies between the genus abundance of saliva microbiota and disease progression during HIV infection. The comparison in Fig. 5A showed that salivary microbial abundances in uninfected individuals were significantly different from patients during AHI (stage 0 group); the abundances of eight genera decreased, whereas those of 34 genera increased. A comparison of the abundance of bacteria from acute to asymptomatic chronic stage (stage 1 group) is shown in Fig. 5B. In the stage 1 group with mild immunosuppression, 10 bacterial genera had significantly changed; the abundances of nine bacteria decreased, whereas that of one genus increased. With the decrease in CD4 counts, compared with stage 1, the salivary microbial abundance of the patients was altered slightly, and only two bacterial genera changed; both showed decreased abundances in the stage 2 group with moderate immune suppression (Fig. 5C). After entering AIDS stage (stage 3 group), with severe immunosuppression, compared with the stage 2 group, we found that the number of bacteria with significantly altered abundance was 11; the abundances of 10 bacterial genera decreased, whereas that of one bacterial genus increased (Fig. 5D).

3.4. Relationships between CD4 counts, BVLs and HIV-positive salivary microbiota

Correlation heatmaps (Fig. 6) revealed the effects of CD4 counts and BVLs on HIV-positive salivary microbial communities. The elevated viremia in untreated patients was associated with a decrease in the proportions of *Haemophilus* and *Leptotrichia* and was more significantly negatively related to *Capnocytophaga* and *Fusobacterium*. We found that *Haemophilus* was not only negatively correlated with BVL but also positively correlated with CD4, whereas *Solobacterium* increased in abundance as CD4 counts decreased (r: −0.2984, $P = 0.04913$). The details were shown in Table 3.

4. Discussion

This study was the first to perform a clear stage comparison of salivary microbial differences in HIV-infected patients. Based on the inflammatory characteristics of progression and a previous study describing salivary diversity [25], we made similar predictions, assuming that the alpha-diversity of infected saliva would decrease significantly and show some volatility with inflammation and immunosuppression. However, the results were opposite our predictions, it was found that oral microbiome changes in the saliva of untreated HIV-positive patients, except for those with AIDS (stage 3), were characterized by greater alpha-diversity (richness and diversity), with stage 2 having the largest difference from the control group, which is similar to the fecal microbiota in untreated PLWHA [28]. Additionally, PLS-DA of beta-diversity could clearly distinguish HIV-positive and -negative samples. HIV-positive salivary microbiota gradually migrated from the stage 0 to stage 3 group, and these microbiota were clearly different from those in the negative control group. The findings are indicative of microbiome alteration during acute HIV infection that persisted over time, with results similar to the changes in the intestinal mucosa [28, 29]. Based on previous reasoning and immunocompromised conditions, the alpha-diversity in stage 3 should be significantly higher than that in controls owing to the interactions of severe immunosuppression and the microbiome [30]. However, in this study it was observed that although the inclusion of three patients with oral candidiasis made the analysis of the oral microbial environment more complicated, patients with AIDS had lower OTU levels in the salivary microbiota during oral inflammation which was in line with a previous study [17]. Another report suggested that there may be a positive symbiotic relationship between bacteria and fungi in the asymptomatic oral cavity [8], although inflammatory factors greatly affect the diversity of the saliva. However, further studies are required to determine whether symbiosis still exists between oral bacteria and fungi in the context of fungal infection.

In the negative controls, eight genera (*Streptococcus*, *Porphyromonas*, *Prevotella_7*, *Prevotella*, *Rothia*, *Neisseria*, *Actinomyces* and *Veillonella*) which accounted for more than 70% among the 175 genera were large proportion genera in the four positive groups, which were described as “core microbiome” [31]. Data analysis was based on a lack of HIV oral specific lesions (except for oral candidiasis); therefore, the similarities in the main genera could be acceptable. Notably, *Porphyromonas* had an absolute advantage in the control group, much higher than its proportion in the other positive groups, and these results were supported by a previous study [25]. However, another report revealed an overgrowth of *Porphyromonas* in the untreated PLWHA [22]. This discrepancy might be due to the latter design that was a longitudinal study without comparing

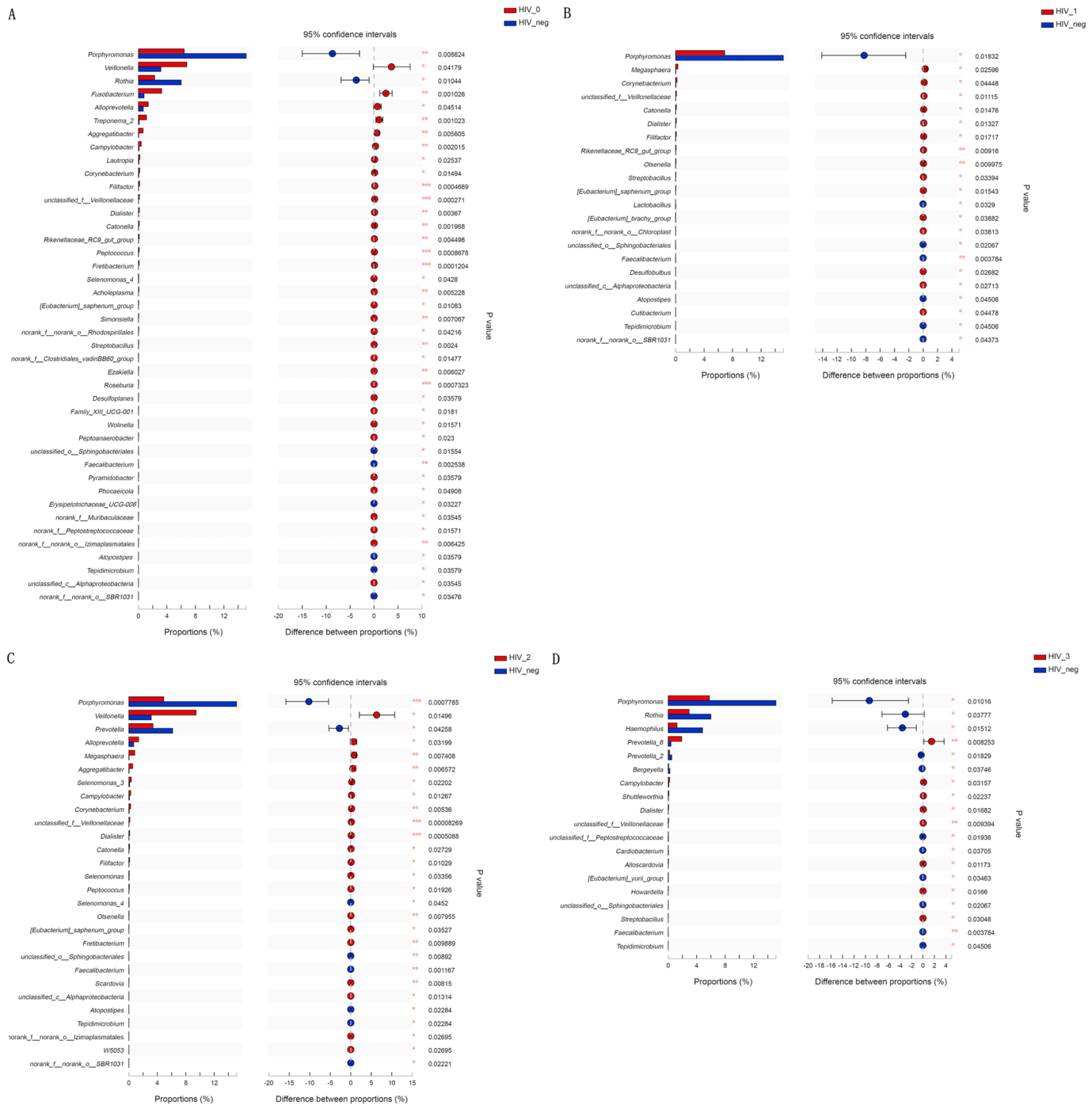


Fig. 3. Differences in the abundance data at the genus level between HIV-positive groups and HIV-negative controls. Significance was evaluated using Wilcoxon rank sum tests. *$0.01 < P \leq 0.05$, **$0.001 < P \leq 0.01$, ***$P \leq 0.001$. A. Stage 0 group and HIV-negative controls. B. Stage 1 group and HIV-negative controls. C. Stage 2 group and HIV-negative controls. D. Stage 3 group and HIV-negative controls.

uninfected controls. Importantly, the reduced relative abundance of *Porphyromonas* may be related to the invasion of HIV, which causes the body to produce a series of immune responses. HIV specifically binds to *Porphyromonas*, causing the virus to enter epithelial cells through a receptor-independent pathway, affecting the oral microbial community [32].

The relative abundance of the saliva microbiota at the genus level is related to the progression of HIV disease. Once HIV invades the human body, under different CD4 counts, the human saliva microbiota varies continuously. Compared with the negative control group, this study revealed that the abundances of 34 genera increased abruptly in the acute stage, whereas five (*Treponema_2*, *Lautropia*, *Fretibacterium*, *Roseburia*, and *Pyramidobacter*) began to decrease in abundance in stage 1. During stage 2, only two genera of bacteria differed in abundance compared with that in stage 1 and showed a significant decrease. There were fewer differences during the asymptomatic stage. Clinical manifestations became increasingly apparent as CD4 counts declined at stage 3, and the abundances of 10 bacterial genera in stage 2 further decreased. According to an overall analysis of the stages of the disease, the salivary microbiota changed dramatically during the acute infection stage owing to the strong pro-inflammatory cytokine storm [33]; symbiosis changed to dysbiosis, and there was increase in the proportions of

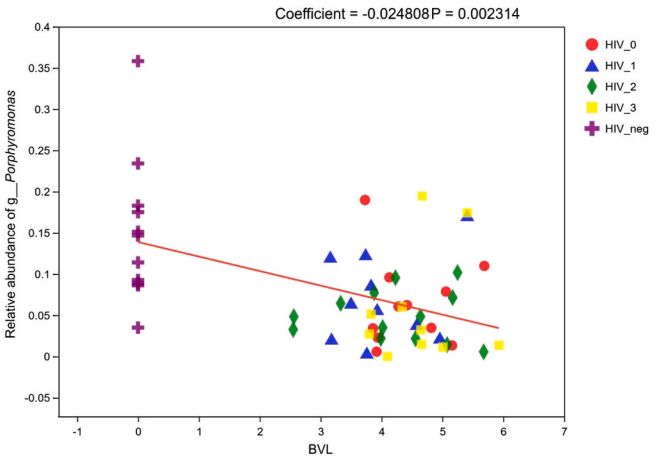


Fig. 4. Multivariate Association with Linear Models (MaAslin) analysis shows the correlation between BVL and the relative abundance of *Porphyromonas*. Coefficient is the correlation coefficient between BVL and *Porphyromonas* (>0 positive correlation, <0 negative correlation, = 0 no correlation), $P < 0.05$ means significant correlation.

previously negligible bacteria in the oral cavity [14]. When HIV antibodies appeared to antagonize the virus, some of the abnormally proliferating microbiota attempted to adjust to a steady state; however, the saliva microbiota was still unbalanced compared with the control group. In the context of mild and moderate immunosuppression, the salivary microbial environment was similar, although the imbalance in the latter was more severe. It is still unclear whether changes in the saliva microbiome were the same before and during AIDS opportunistic oral infections. However, the results definitely confirmed that the suppression of a few common species could lead to increased risk of disease.

Saliva bacterial communities in our HIV-positive MSM cohort were affected by CD4 counts and BVLs. In this study, *Haemophilus*, *Leptotrichia*, *Capnocytophaga* and *Fusobacterium* of the resident oral microbiota were negatively correlated with HIV viral load and *Haemophilus* was positively correlated with CD4; Additionally, other reports revealed the four genera had been implicated in opportunistic infections of HIV-positive individuals [21,34–36]. In view of this result, the study claims that regardless of the low BVL or the high CD4 value, untreated PLWHA has the potential risk of oral opportunistic infection. Interestingly, none of the bacteria with a high abundance ratio increased in abundance as the viremia progressed, although the above four opportunistic pathogens showed decreased abundance, which was very different from the microbiological studies of the tongue mucosa [12], potentially because of the small sample size of the latter and different controls. Meanwhile, the overall bacterial population in the saliva was affected as a result of viral replication, especially *Solobacterium*, this in line with the analysis of the tongue mucosa [12]. Moreover, owing to the unique negative relationship between the CD4 count and *Solobacterium*, the role of *Solobacterium* should be studied further. *Solobacterium* is a gut bacterium, it can be found in patients with oral diseases and immunodeficiency [37]. Anal and oral sex are the main methods for sexual intercourse in MSM [38,39]. Microbiota in the gut can easily transfer to the oral cavity and cause proliferation. Recent studies have confirmed that halitosis [40–43], endodontic abscesses [44], periodontal disease [45] and oral carcinoma [46] are associated with abundant *Solobacterium* in the oral cavity. *Solobacterium* showed enhanced growth as CD4 decreased, and exhibited enhanced ability to induce inflammation and drive oral lesions.

There were several limitations to our study. First, the number of patients in our study cohort was low, and the MSM in our cohort may not reflect the overall PLWHA population in China. Second, this study was a cross-sectional study. Because of ethical considerations, we could not

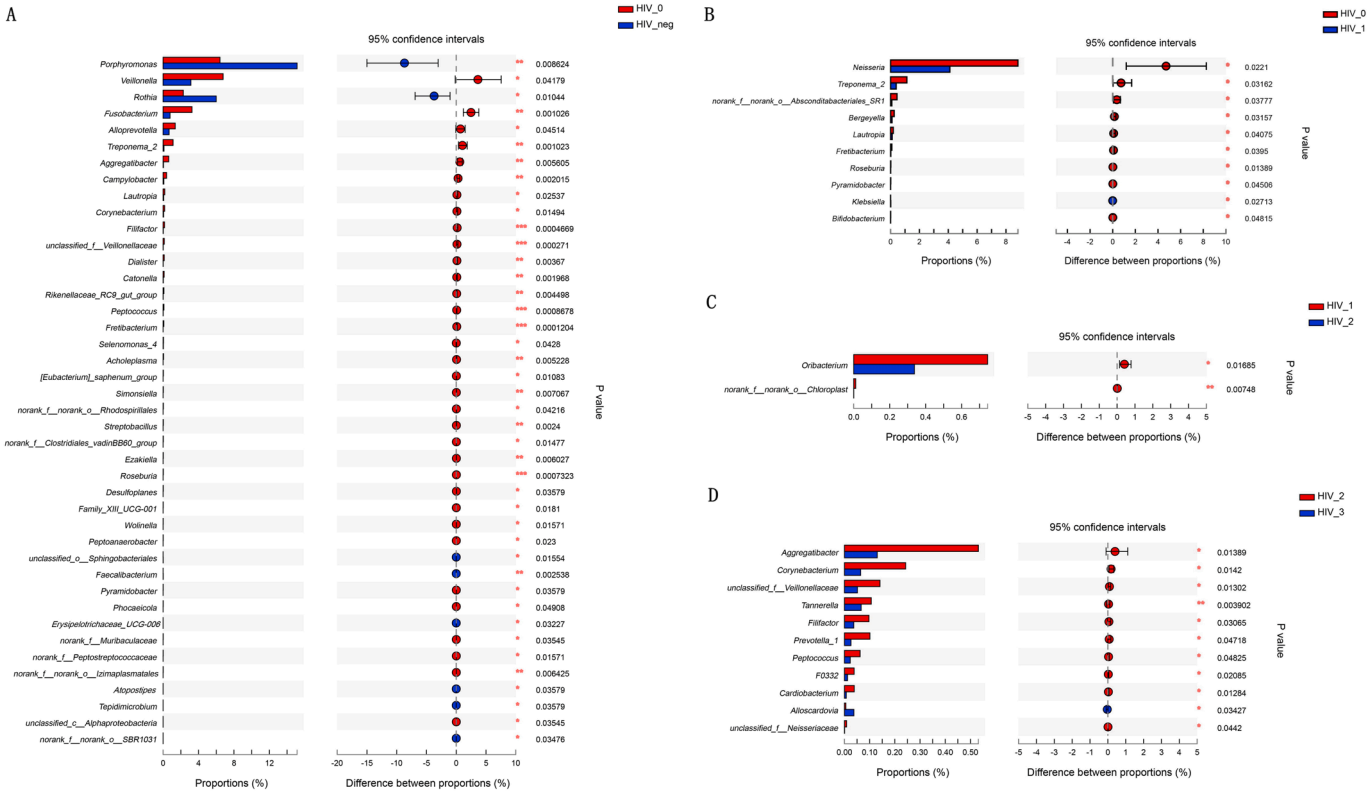


Fig. 5. Bar plots were obtained by applying Wilcoxon rank sum tests, and the abundances of genera were compared among HIV-positive groups according to disease progression. *$0.01 < P \leq 0.05$, **$0.001 < P \leq 0.01$, ***$P \leq 0.001$. A. HIV-negative controls and Stage 0 group. B. Stage 0 group and Stage 1 group. C. Stage 1 group and Stage 2 group. D. Stage 2 group and Stage 3 group.

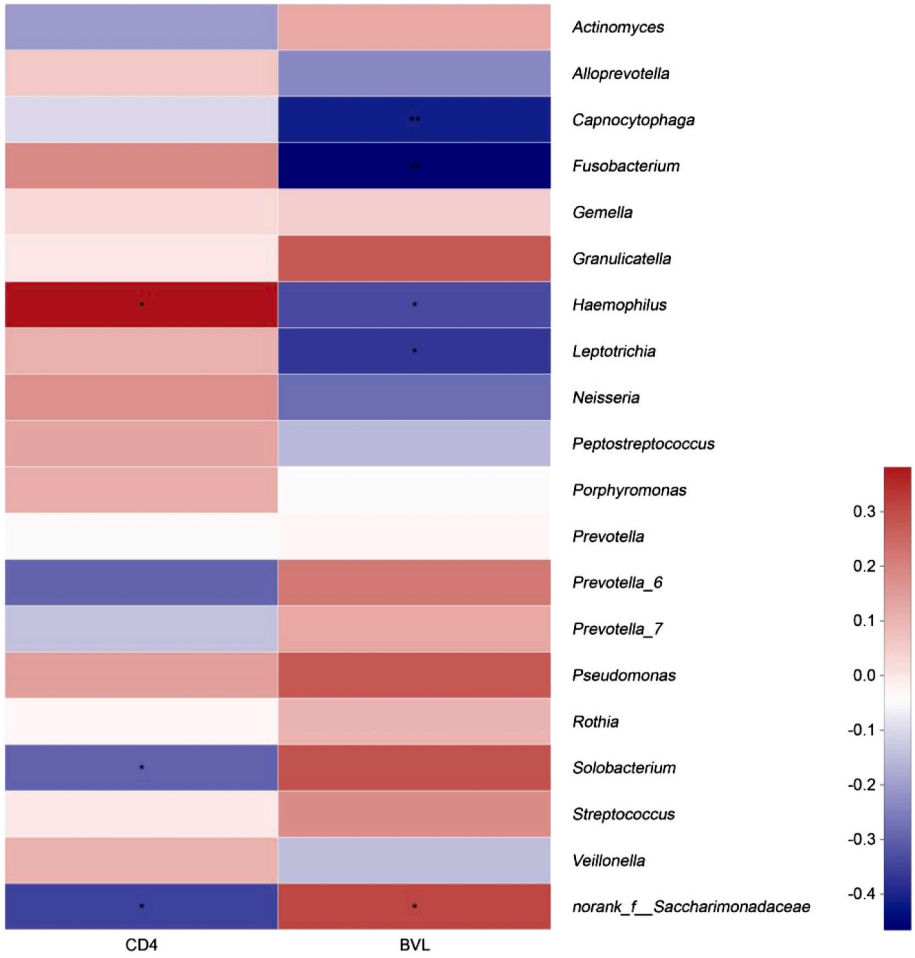


Fig. 6. Correlation heatmap analysis was used to visualize the relationships among the top 20 most abundant salivary microbiota, CD4 counts and BVL variables on HIV-positive salivary microbial communities. R values (Spearman correlation coefficient) are shown using different colors in the figure; red color represents positive correlation, and blue color represents negative correlation. *$0.01 < P \leq 0.05$, **$0.001 < P \leq 0.01$.

Table 3
Correlation analysis between salivary microbiota and CD4 or BVL in HIV-positive groups.

Genus	*Capnocytophaga*		*Fusobacterium*		*Haemophilus*		*Leptotrichia*		*Solobacterium*	
	r value	*P* value	*r* value	*P* value	*r* value	P value	*r* value	*P* value	*r* value	*P* value
CD4	−0.10078	0.51511	0.18562	0.22768	0.38154	0.01061*	0.11121	0.47236	−0.2984	0.04913*
BVL	−0.41139	0.00553**	−0.46585	0.00144**	−0.33561	0.02594*	−0.36888	0.01374*	0.2918	0.05462

Statistical analysis was performed by Spearman rank correlation analysis. R value (Spearman correlation coefficient): $r > 0$ represents positive correlation, $r < 0$ represents negative correlation and $r = 0$ represents no correlation; *$0.01 < P \leq 0.05$, **$0.001 < P \leq 0.01$.

compare the oral microbiota of the same untreated patient at different stages of infection to explore the relationships between immune status and oral microbial development. Additionally, because of the many factors affecting microbiological research, it is challenging to design experiments for longitudinal self-comparison for the entire HIV infection process. Finally, the lack of strict oral environmental controls in the MSM population, such as smoking, periodontal disease, and dental caries, could lead to biased results.

5. Conclusion

In summary, PLWHA at different disease stages exhibited different saliva microbial features. With the progression from acute to asymptomatic stages, the richness and diversity of OTUs increased, whereas abnormal proliferation of non-dominant bacteria decreased. AIDS stage showed different alterations, including a lower OTU level and suppression of non-dominant bacteria. Drawback of this study is the small sample size that could have contributed to bias, and some characteristic genera could have been omitted. However, changes in the characteristic genera of the saliva were related to inflammation induction and immune activation. Our findings also revealed the relationship between untreated PLWHA salivary microbiota and immunosuppressive status, particularly the characteristic changes in the salivary microbiota in the acute and AIDS stages, which demonstrated the potential risk of oral infection in the seemingly healthy PLWHA. Further research will aid in evaluating the effects of the features and functions of *Porphyromonas* and *Solobacterium* on the PLWHA at different stages, in order to improve our understanding of the oral microenvironment before immune reconstruction under different initial immune states.

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Ethics approval

The protocol for all procedures was approved by the Institutional Review Board of Beijing Youan Hospital, Capital Medical University. The ethical approval number is Jingyou Kelun Zi [2017] No.30. All participants signed a committee-approved informed consent form.

CRediT authorship contribution statement

Ying Guo: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Wei Xia:** Investigation, Resources. **Feili Wei:** Conceptualization, Formal analysis, Resources. **Wei Feng:** Investigation, Resources. **Junyi Duan:** Investigation, Resources. **Xintong Sun:** Investigation, Data curation. **Yixi Yu:** Investigation, Data curation. **Yan Wang:** Investigation, Data curation. **Baojin Zhang:** Investigation, Data curation. **Jie Cao:** Investigation, Data curation. **Weidi Li:** Investigation, Resources. **Haozhi Dong:** Investigation. **Shan Guo:** Resources. **Fengqiu Zhang:** Conceptualization, Data curation, Writing – review & editing, Supervision. **Xiaojie Huang:** Conceptualization, Formal analysis, Resources, Writing – review & editing, Project administration, Funding acquisition. **Zheng Sun:** Conceptualization, Methodology, Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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